Cadence Modulation during Eccentric Cycling Affects Perception of Effort But Not Neuromuscular Alterations

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ABSTRACT

MATER, A., A. BOLY, A. MARTIN, and R. LEPERS. Cadence Modulation during Eccentric Cycling Affects Perception of Effort But Not Neuromuscular Alterations. *Med. Sci. Sports Exerc.*, Vol. 56, No. 5, pp. 893–901, 2024. **Introduction:** A recent study showed that cadence modulation during short eccentric cycling exercise affects oxygen consumption ($\dot{V}O_2$), muscular activity (EMG), and perception of effort (PE). This study examined the effect of cadence on $\dot{V}O_2$, EMG, and PE during prolonged eccentric cycling and exercise-induced neuromuscular alterations. **Methods:** Twenty-two participants completed three sessions 2–3 wk apart: 1) determination of the maximal concentric peak power output, familiarization with eccentric cycling at two cadences (30 and 60 rpm at 60% peak power output), and neuromuscular testing procedure; 2) and 3) 30 min of eccentric cycling exercise at a cadence of 30 or 60 rpm. PE, cardiorespiratory parameters, and vastus lateralis and rectus femoris EMG were collected during exercise. The knee extensors' maximal voluntary contraction torque, the torque evoked by double stimulations at 100 Hz (Dt100) and 10 Hz (Dt10), and the voluntary activation level were evaluated before and after exercise. **Results:** $\dot{V}O_2$, EMG, and PE were greater at 30 than 60 rpm (all $P < 0.05$). Maximal voluntary contraction torque, evoked torque, and Dt10/Dt100 ratio decreased (all $P < 0.01$) without cadence effect (all $P > 0.28$). Voluntary activation level remained constant after both eccentric cycling exercises ($P = 0.87$). **Conclusions:** When performed at the same power output, eccentric cycling exercise at 30 rpm elicited a greater PE, EMG, and cardiorespiratory demands than pedaling at 60 rpm. Exercise-induced fatigability was similar in both eccentric cycling conditions without neural impairments, suggesting that eccentric cycling seemed to alter more specifically muscular function, such as the excitation–contraction coupling process. In a rehabilitation context, eccentric cycling at 60 rpm seems more appropriate because it will induce lower PE for similar strength loss compared with 30 rpm. **Key Words:** NEUROMUSCULAR ALTERATION, ECCENTRIC CYCLING CADENCE, MUSCULAR ACTIVITY, PERCEIVED EFFORT, OXYGEN CONSUMPTION

Rehabilitation programs for patients suffering from cardiorespiratory diseases such as chronic obstructive pulmonary disease aim to maintain quality of life and reverse the deleterious impact caused by loss of muscle mass and the progressive worsening of dyspnea. Rehabilitation programs incorporate locomotor exercises such as walking, running, or cycling, as they involve large muscle mass needing a certain amount of cardiorespiratory demand. According to the ventilatory constraints and the lower limb muscle weakness of these patients, eccentric (ECC) cycling has been proposed as a locomotor activity (1) allowing to maximize benefits from rehabilitation (2). Indeed, ECC cycling requires 2.5 to 6 times less oxygen consumption required in concentric cycling at the same power output (3). Another benefit of ECC cycling is that it is perceived as less difficult (4). Even with these advantages, patients who show increased fatigability are not guaranteed to adhere to this type of exercise (5), and attention must be taken to how a session is prescribed to control exercise-induced fatigue. Interestingly, compared with concentric cycling, 30 min of ECC cycling performed at a similar power output induces a similar loss of strength (4,6). Moreover, even if ECC cycling can induce muscle damage (6), ECC-induced muscle damage can be reduced by increasing the exercise intensity progressively during training programs (7).

Still, for a given power output, a wide selection of cadences can be used. The only recommendations from previous studies (8,9) examining ECC cycling in patients proposed to use low cadence (i.e., 15 rpm) according to an older investigation assessing alterations after ECC maximal voluntary contraction (MVC) performed with biceps brachii (BB) (10). Indeed, this study suggested that slow muscle contractions lead to lower muscle soreness compared with greater muscle velocity contraction. However, for a given power output, increasing the cadence results in a decrease in the torque applied to the pedals, and vice versa. Furthermore, it has been demonstrated that physiological responses are modulated by cadence in concentric pedaling. In a context of ECC cycling, a previous study conducted in our laboratory showed that, for a given power

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output, increasing or decreasing cadence from optimal ones led to increase physiological and perceptual responses. Thus, used a cadence of 30 rpm generated greater perceived effort, oxygen consumption, and muscular activity of knee extensors compared with pedaling at 60 rpm (11). It is possible that these acute modulations of the physiological functions could lead to further impacts on fatigability and perceived effort when exercise lasts over time. Only one study (12) compared the effect of maximal ECC single-leg cycling exercise at 5 and 35 rpm on performance fatigability. Concentric knee extensors MVC torque (measured at $30°\cdot s^{-1}$ and $210°\cdot s^{-1}$) decreased after cycling at 35 rpm, associated with a power output five times greater during the condition at 35 rpm. Regarding markers of muscle damage, delayed-onset muscle soreness was greater until 4 d after cycling exercise at the higher cadence. In contrast, other markers of muscle damage, such as joint flexibility or muscle swelling, did not differ between pedaling cadence conditions. A recent study by the same team of researchers complements these results. The condition at 35 rpm was kept the same, and in a second condition, participants had to maintain maximal ECC pedaling exercise at 5 rpm until they had completed the same amount of work. They showed that the exercise at 5 rpm lasted five times longer than the condition at 35 rpm, but in this context, the decrease in MVC torques, as well as the markers of muscle damage, was similar between the two cadences used (13). Unfortunately, a 5-min or prolonged maximal exercise used in those studies (12,13) did not reflect the rehabilitation program for patients that generally lasts around 30 min at submaximal intensity.

In this context, the present study aimed to compare i) the perceptual and cardiorespiratory responses during a 30-min ECC cycling exercise performed at two different cadences (30 vs 60 rpm) and ii) the changes in neuromuscular function of the knee extensor muscles after exercise. We hypothesized that muscular activation, perception of effort, and cardiorespiratory responses would be greater during ECC cycling exercise at 30 rpm compared with 60 rpm, as demonstrated in a previous study (11), but neuromuscular alterations would be similar for both cadence conditions

METHODS

Participants. A required sample size of 16 was calculated based on the loss of MVC force led by cadences of 30 min of concentric cycling at 50 and 110 rpm (14); from a dependent means *t*-test, an effect size of $d = 0.66$—calculated from presented data—an α level of 0.05, and a power 0.80. Then, 22 healthy volunteers (13 men and 9 women) participated in this study (age, 22 ± 3 yr; height, 168 ± 19 cm; and body mass, 68.5 ± 7.7 kg). Most participants were novices to ECC cycling and only practiced concentric cycling for daily commutes. Only three of them participated in an ECC cycling experiment 1 yr before the present one. Participants were free from injuries in their lower limbs during the last 3 months. This study was conducted in accordance with the Declaration of Helsinki (2008), and all participants signed a written informed consent. The experimental design of the study was approved by the regional ethics committee (CPP EST: approval no. A00064-49). Their mean concentric cycling peak power output (PPO) was 235.5 ± 46.7 W, corresponding to a peak oxygen uptake ($\dot{V}O_{2peak}$) of 50.9 ± 7.8 $mL\cdot kg^{-1}\cdot min^{-1}$ and a maximal heart rate (HR) of 183 ± 9 bpm. Participants were thus defined as recreationally trained (15,16).

Experimental procedures. All participants completed one familiarization session and two experimental sessions, separated by 3 wk to 1 month. Experimental sessions were completed in randomized order at the same time of day (±1 h). Participants were instructed to avoid fatiguing exercises during the 2 d before each session and to keep the usual diet. During the familiarization session, the volunteers realized a continuous incremental cycling test (starting at 50 W with an increase of 1 W/3 s) on the concentric ergometer until exhaustion at freely chosen cadence, allowing to determine PPO (as the greater achieved power output before exhaustion) and maximal cardiorespiratory function. Then, the assessment of the knee extensors by neuromuscular function was realized before and immediately after 30 min of ECC pedaling (15 min per cadence, 30 and 60 rpm in randomized order) at 60% PPO. This session aimed to familiarize the volunteers with the experimental procedures. Two semirecumbent cycle ergometers were used for concentric (Ergoline GmbH, Ergoselect 600) and ECC (Cyclus 2, Cyclus GmbH) cycling. The seat of the bikes was adjusted for each volunteer to have the same knee angles between the two bikes. Each experimental session started with a 10-min warm-up at 40% PPO followed by 30 s at PPO in concentric cycling. Then, participants completed two 30-min sessions of ECC cycling at 60% PPO in isoload mode: one at 30 rpm (ECC30) and one at 60 rpm (ECC60), preceded and followed by an assessment of neuromuscular function of knee extensors. Between the end of the cycling exercise and the beginning of the neuromuscular function tests, the time latency was 1 min. Cardiorespiratory variables and muscular activity (EMG) were assessed during the pedaling exercise, and the participants rated their perceived exertion and muscle pain every 2 min in random order.

Electromyography. Bipolar wireless EMG units (Pico Cometa; Biometrics, Gometz-le-Châtel, France) were fixed on Ag/AgCl electrodes (recording over 10 mm; a center-to-center distance of 15 mm—30 × 24-mm ECG Electrode, Kendal; Cardinal Health, Dublin, OH). The skin of the participants was prepared (shaved and cleaned with alcohol swabs), and electrodes were positioned on the belly of the vastus lateralis (VL), rectus femoris (RF), and BB muscles on the right part of the body, following the recommendations of Barbero et al. (17). The first session allowed us to define electrode positions using anatomical landmarks for all sessions. During the neuromuscular assessments and cycling exercises, EMG and torque were recorded at 2000 Hz with a Biopac MP150 unit and stored on AcqKnowledge 4.2 software (Biopac Systems Inc., Goleta, CA) for offline analysis. EMG signals were amplified by 200 and filtered between 10 and 390 Hz

Neuromuscular function. Volunteers were seated on an isokinetic dynamometer (System pro 5; Biodex Medical System, Shirley, NY) with a hip and knee flexion at an angle of 90° (with an angle of 0° corresponding to maximal extension). Their trunk was maintained against the chair back with a belt to avoid any body movement. The knee angle axis was placed in front of the rotation axis dynamometer, and the right leg was strapped at the level of the ankle, using the articulated arm, 2 cm above the internal malleolus.

Percutaneous electrical stimulations were applied using a high-voltage constant–current stimulator (model DS7AH; Digitimer, Fort Lauderdale, FL) via a ball probe cathode (0.5-cm diameter), covered with cotton to reduce pain and manually pressed by the experimenter on the femoral nerve using a stylus. A self-adhesive anode (8 × 4 cm) was placed in the gluteal fossa. Stimulation intensity was progressively increased until both the evoked torque and the M-wave responses of VL and RF muscles plateaued. We then used 150% of this intensity (500 μs, 400 V) to assess maximal M-wave (M_{MAX}) and associated muscle twitch properties.

Participants performed MVCs of the knee extensors with their right leg during which there were strongly encouraged. Two MVCs were required (or three if the peak torque difference was greater than 5%) before the cycling exercise. The higher trials obtained before exercise were analyzed. After the exercise, the neuromuscular testing procedure was performed at least one time as fast as possible after the end of the exercise. During MVCs, one double stimulation (doublet (Dt)) at 100 Hz was manually delivered, and immediately after, at rest, two doublets (respectively at 100 and 10 Hz) as well as a single twitch (Tw) were automatically sent with a 4-s delay between them. The site of stimulation was marked for postexercise stimulations.

Cardiorespiratory responses during the exercises. The wearable metabolic system (K5; Cosmed, Brignais, France) was calibrated before each session according to company guidelines. The mask associated with the gas analyzer was fixed on the face of the participants just before the start and removed directly after the end of the cycling exercise. Maximal oxygen uptake and HR were assessed in the last 15 s of the incremental test. Preexercise and postexercise neuromuscular testing were not performed with the mask to avoid impairing the procedure. The oxygen uptake ($\dot{V}O_2$; $mL \cdot kg^{-1} \cdot min^{-1}$), respiratory frequency (bpm), and minute ventilation ($\dot{V}E$, $L \cdot min^{-1}$) were recorded during exercise. The HR (bpm) was obtained using a chest strap HR monitor (HR Module; Garmin, Olathe, KS) paired with the metabolic system.

Perceptual parameters. Participants were familiarized with the Borg CR100 Scale to measure perceptual parameters during the first session across the incremental test and ECC pedaling. They announced their perception of effort (i.e., difficulty to control their legs and to breathe during cycling exercises) and muscle pain (i.e., how much it globally hurt in their leg muscles during cycling exercise). The two perceptual parameters were reported every 2 min during ECC pedaling in random order. In addition, delayed-onset muscle soreness (assessed by the question: what is your overall leg muscle pain when standing up from a chair?) was reported 24, 48, and 72 h after the end of the cycling exercises.

Data analysis. Data 2.5 times SDs below and above the mean value were considered outliers and excluded from our statistical analysis. It represented 1.84% of data and was not kept for statistical analysis or presented in graphs or figures. For one participant, the double stimulations were not tolerated, and for another, the adhesive anode did not receive current after exercise for one session because of excessive sudation.

During pedaling exercise, metabolic parameters, subjective feeling, and muscular activity were averaged every 6 min during cycling exercises (five periods of time). Oxygen uptake and HR were expressed as a percentage of peak oxygen uptake obtained during the concentric incremental test and averaged over the last 15 s. Respiratory exchange ratio was average over the duration of exercise. EMG root mean square (RMS) was normalized by the EMG RMS obtained during the 30 s at PPO_{CONC} after the warming up of the same session. Perception of muscle pain sitting on the ergometer before the start of pedaling exercise was subtracted from muscle pain while pedaling. Maximal torque amplitude was analyzed for Dt100, Dt10, single twitch, and MVC before and after cycling exercises. Dt10/Dt100 ratio was used to indicate a change in excitation–contraction coupling (18). The contraction time (Ct) of Dt100, Dt10, and single twitch was measured from the onset of torque increase to its peak value. The rate of torque development (RTD; $N \cdot s^{-1}$) was calculated by dividing the peak torque value by Ct, whereas the half-relaxation rate (HRR; $N \cdot s^{-1}$) was measured from the end of the peak torque plateau—when applicable—to the point it fell to half its peak value. Both variables were the ratio between changes of torque upon the time. Strojnik and Komi's formula was used to obtain the voluntary activation level (VAL) (19), which includes the torque values at the moment of the stimulation (MVCstim) and added by the stimulation (Dtsup):

$$\mathrm{VAL} = \left[1 - \left(\left(\frac{\mathrm{MVCstim}}{\mathrm{MVC}} \right) \times \left(\frac{\mathrm{Dtsup}}{\mathrm{Dt100}} \right) \right) * 100 \right]$$

M_{MAX} duration, amplitude, and area for RF and VL muscles were investigated. RMS EMG during MVC was measured during 500 ms around the peak force and divided by M_{MAX} amplitude (EMG_{MAX}) to estimate the magnitude of voluntary neural drive. All data are expressed as mean ± SD. Minimal and maximal values among the participants were added to Table 2.

Statistical analyses. According to our hypothesis, data from ECC30 were compared with the ECC60. The effect of cadence over the exercise duration on perceptive parameters, muscular activity, and cardiorespiratory function was assessed using two-way repeated-measures ANOVAs (CADENCE (2) × TIME (5)). Two-way repeated-measures ANOVAs (CADENCE (2) × TIME (2)) was used to compare neuromuscular function before and after ECC cycling exercises. Last two-way repeated-measures ANOVAs (CADENCE (2) × TIME (2)) were used to evaluate the onset of perceived muscle pain days after the exercise compared with preexercise values (CADENCE (2) × TIME (4)). ANOVAs were used because all data followed a normal distribution (Shapiro–Wilk test). In case sphericity was violated, correction of Greenhouse–Geisser was applied.

APPLIED SCIENCES

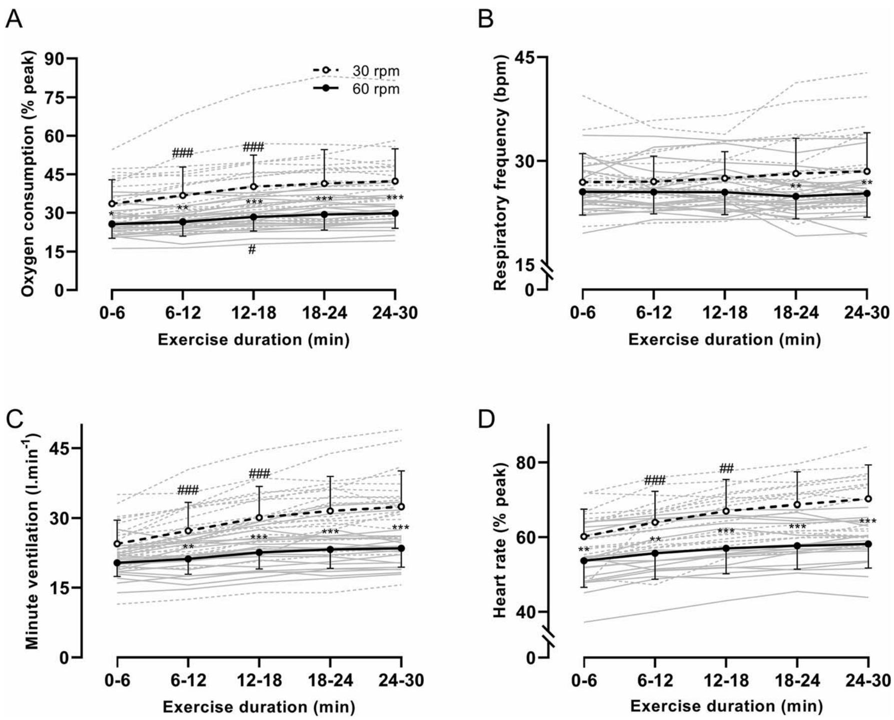

FIGURE 1—Cardiorespiratory function during the cycling exercises. Gray lines represent individuals. Panels A, B, C, and D represent oxygen consumption, respiratory frequency, minute ventilation, and HR, respectively. Signs are placed below the line for the 60-rpm condition and above for the 30-rpm condition, and between both lines when *post-hoc* test shows a difference between 30 and 60 rpm. # indicates an increase of responses compared with the preceding 6 min, * means that responses were greater at 30 rpm compared with 60 rpm at the same time interval. For every symbol, one sign indicates $P < 0.05$, two signs indicate $P < 0.01$, and three signs indicate $P < 0.001$. Error bars represent the SD.

When significance was reached, *post hoc* multiple comparisons using the Holm procedure were conducted. In all statistical tests, the level of significance was set at $P < 0.05$. Effect sizes (dz) and partial eta square (η_p^2) were calculated for followed-up analyses. Repeated-measures correlation coefficients (r_{rm}) were calculated between perception of effort and its physiological correlates for both conditions together and separately at five time points. In parallel, repeated-measures correlation coefficient between oxygen consumption and EMG activity of the VL and RF muscle was determined. Statistical analysis was performed with the software JASP (version 0.16.1; JASP TEAM, Amsterdam, the Netherlands) and R project (version 4.1.3; R Core Team).

RESULTS

Responses during the exercise. All cardiorespiratory parameters showed a CADENCE × TIME interaction (all $P < 0.01$; all $\eta_p^2 > 0.20$; Fig. 1). Oxygen consumption was greater during pedaling at 30 rpm compared with 60 rpm. HR and minute ventilation showed a similar pattern as oxygen consumption. Respiratory frequency remained unchanged for both cadences until 18 min of exercise, then was greater at 30 rpm compared with 60 rpm. Respiratory exchange ratio (0.76 ± 0.05) did not differ between both condition ($P = 0.89$; $\eta_p^2 = 0.00$).

EMG activity of VL and RF muscles is presented in Figure 2, panels A and B, respectively. VL muscle activity showed a

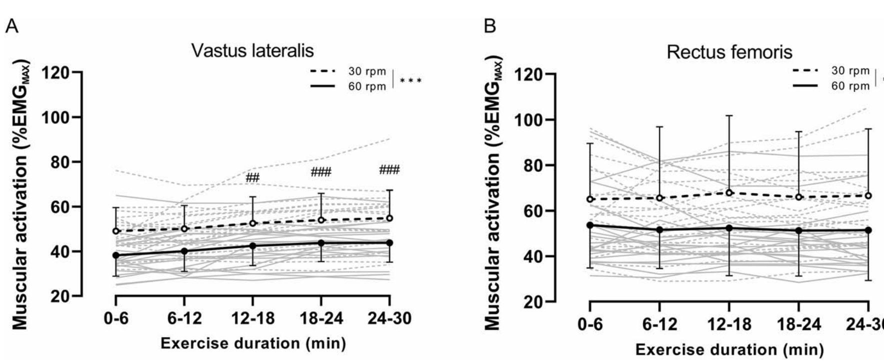

FIGURE 2—Mean (SD) of muscular activity of the different lower limb muscles. Panels A and B represent VL and RF muscles, respectively. Both VL and RF muscles showed a main effect of CADENCE. Gray lines represent individuals. # indicates an increase in responses compared with the preceding 6 min for both conditions.* indicates difference between both conditions. For every symbol, one sign indicates $P < 0.05$, two signs indicate $P < 0.01$, and three signs indicate $P < 0.001$. Error bars represent the SD.

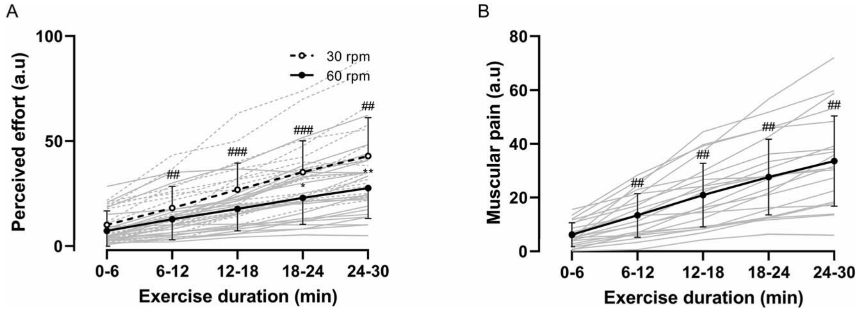

FIGURE 3—Mean (SD) of perceived effort and lower limb muscle pain. Panels A and B represent the perception of effort and muscle pain, respectively. Values of the two cadences are averaged when two-way repeated-measures ANOVA did not show a significant CADENCE × TIME interaction effect. Gray lines represent individuals. Signs are placed below the line for the 60-rpm condition and above for 30-rpm condition, and between both lines when *post-hoc* test shows a difference between 30 and 60 rpm. # indicates an increase of responses compared with the preceding 6 min. * means that responses were greater at 30 rpm compared with 60 rpm at the same time interval. For every symbol, one sign indicates $P < 0.05$, two signs indicate $P < 0.01$, and three signs indicate $P < 0.001$.

main effect of CADENCE ($P < 0.001$; $\eta_p^2 = 0.517$) with greater response at 30 rpm than at 60 rpm, and also increased with TIME ($P < 0.001$; $\eta_p^2 = 0.469$). RF muscle activity showed only a main effect of CADENCE ($P < 0.05$; $\eta_p^2 = 0.268$). EMG activity of BB muscle showed a CADENCE × TIME interaction ($P < 0.05$; $\eta_p^2 = 0.19$) and a main effect of cadence ($P < 0.01$; $\eta_p^2 = 0.34$). BB muscle activity was greater at the three last time periods than the second one only at 30 rpm.

Perception of effort showed a CADENCE × TIME interaction ($P < 0.001$; $\eta_p^2 = 0.39$; Fig. 3A). Precisely, perception of effort increased in both conditions and was greater at ECC30 compared with ECC60. As shown in Figure 3B, perceived muscle pain increased with TIME ($P < 0.001$; $\eta_p^2 = 0.74$) without difference between cadence ($P = 0.14$; $\eta_p^2 = 0.1$). It increased at each duration point compared with the preceding 6 min (all $P < 0.01$; all $\eta_p^2 > 0.42$).

Repeated-measures correlations. The repeated-measures correlation coefficients between perceived exertion and its correlates, and between oxygen consumption and muscle activation of the lower limb muscles are presented in Table 1. All correlations have been presented both grouped and separately for each cadence.

TABLE 1. Repeated-measures correlation (Rmcorr) for correlates of perception of effort and oxygen consumption.

	Correlation with Perception of Effort				Correlation with Oxygen Consumption	
	Respiratory Frequency	EMG RF	EMG VL	EMG BB	EMG RF	EMG VL
Rmcorr both condition						
P	<0.001	<0.001	<0.001	<0.001	<0.001	<0.001
Correlation coefficient	0.38	0.32	0.55	0.32	0.50	0.68
Rmcorr ECC30						
P	0.001	<0.001	<0.001	<0.001	<0.001	<0.001
Correlation coefficient	0.52	0.45	0.59	0.48	0.47	0.74
Rmcorr ECC60						
P	0.22	0.07	<0.001	0.27	0.35	<0.001
Correlation coefficient	−0.13	−0.20	0.62	0.12	0.11	0.68

All correlation coefficients were calculated for five time points per participants.

Neuromuscular changes from before to immediately after the exercise. Knee MVC torque decreased by an average of 17.2% ± 10.1% with TIME ($P < 0.001$; $\eta_p^2 = 0.764$; Fig. 4A) without effect induced by CADENCE ($P = 0.35$; $\eta_p^2 = 0.044$). VAL did not show any effect of TIME or CADENCE (all $P > 0.13$; all $\eta_p^2 < 0.13$; Fig. 4B). Both EMG_{MAX} of VL and RF muscles showed no effect of TIME or CADENCE (all $P > 0.22$; all $\eta_p^2 < 0.11$; Table 2).

Evoked torques (i.e., Dt100, Dt10, Tw; Table 2) decreased by about 22.6% ± 21.1% with TIME (all $P < 0.01$; all $\eta_p^2 > 0.21$) without main effect of CADENCE (all $P > 0.28$; all $\eta_p^2 < 0.07$) except for torque evoked by Dt10 ($P < 0.04$; $\eta_p^2 = 0.214$). Dt10/Dt100 ratio showed TIME and CADENCE effect (all $P < 0.001$; all $\eta_p^2 > 0.532$), without significant CADENCE × TIME interaction ($P = 0.08$; $\eta_p^2 = 0.158$). All Ct, RTD, and HRR associated with electrical stimulations showed a TIME effect (all $P < 0.048$, all $\eta_p^2 > 0.2$) except Dt100 Ct, Tw-RTD, and Tw-HRR.

M-wave characteristics are presented in Table 2. Excepted for RF muscle, M_{MAX} duration decreased with TIME ($P < 0.001$; $\eta_p^2 = 0.537$) whatever CADENCE ($P = 0.87$; $\eta_p^2 = 0.02$), all other M-wave characteristics for both muscles did not show any change (all $P > 0.06$; all $\eta_p^2 < 0.25$).

Delayed-onset muscle soreness. Perceived muscle pain days after exercises showed a main effect of TIME ($P < 0.001$; $\eta_p^2 = 0.31$) only. More precisely, muscle pain increased 24 h after exercise (25.1 ± 10.5 a.u.), compared with preexercise values (1.0 ± 2.1 a.u), remained at high values 48 h after exercise (23.7 ± 14.0 a.u.), and showed a decrease at 72 h after exercise (12.6 ± 10.2 a.u.) but remained higher than value reported before (all $P < 0.001$; all $d > 0.8$, compared with preexercise values).

DISCUSSION

This study was the first to compare exercise-induced fatigue to different cadences (i.e., 30 and 60 rpm) during and after 30 min of ECC cycling at the same power output (60% of concentric PPO). The results showed that perceived exertion, cardiorespiratory responses, and muscular activation were greater during cycling at 30 rpm compared with 60 rpm, but knee

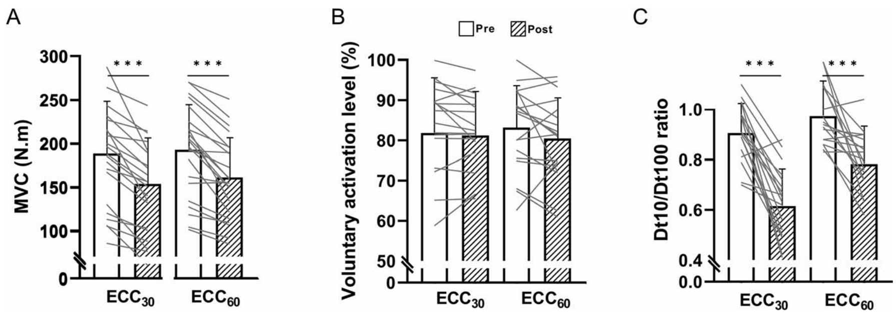

FIGURE 4—Knee extensors maximal voluntary isometric contraction torque, voluntary activation level, and Dt10/Dt100 ratio before and after cycling exercises. Panels A, B, and C represent MVC torque, voluntary activation level, and Dt10:Dt100 ratio, respectively. Gray lines represent individuals. * indicate a difference independent of CADENCE with P < 0.001.**

extensors fatigability did not differ between cadence conditions. Moreover, both conditions lead to similar muscle pain and peripheral function impairment (i.e., evoked torque and excitation–contraction coupling) without central alteration change (i.e., VAL).

During exercise. The present findings on the effect of cadence during ECC cycling on perceptual parameters, cardiorespiratory function, and knee extensor muscle activity complement previous research on ECC cycling. During the initial phase of exercise (0–6 min), cardiorespiratory responses were comparable to those observed in a previous study (11) and were greater at 30 rpm (33.6% $\dot{V}O_{2peak}$) compared with 60 rpm (25.7% $\dot{V}O_{2peak}$). Then, the increase in cardiorespiratory responses was most pronounced at the lowest cadence. The greater cardiorespiratory response at 30 rpm is associated with the higher EMG activity of the VL and RF muscles at

TABLE 2. Changes in neuromuscular parameters from before (PRE) to after (POST cycling exercise.

Variable	Main Effect	Condition	PRE		POST		Effect Size (Dz)
RMS EMG_{MAX}/Mmax	No effect	ECC30	0.048 ± 0.019	(0.023–0.091)	0.049 ± 0.025	(0.024–0.081)	0.28
VL		ECC60	0.064 ± 0.038	(0.025–0.147)	0.069 ± 0.055	(0.026–0.126)	0.05
RMS EMG_{MAX}/Mmax	No effect	ECC30	0.067 ± 0.03	(0.025–0.093)	0.061 ± 0.041	(0.026–0.082)	0.02
RF		ECC60	0.076 ± 0.054	(0.036–0.205)	0.077 ± 0.057	(0.026–0.213)	0.02
Dt100 amplitude	TIME***	ECC30	64.9 ± 16.8	(39.9–90.5)	57.0 ± 19.9	(23.1–89.2)	0.41
(N·m)		ECC60	64.3 ± 18.7	(33.0–98.8)	56.4 ± 20.8	(13.3–99.2)	0.41
Dt100 Ct (ms)	No effect	ECC30	84.0 ± 12.8	(66.5–109.5)	79.1 ± 10.9	(62.5–91.5)	0.34
		ECC60	82.8 ± 14.8	(58.0– 105.0)	78.8 ± 18.0	(60.0–88.0)	0.28
Dt100 RTD	TIME*	ECC30	797 ± 267	(391–1276)	723 ± 242	(302–1067)	0.26
($N·s^{-1}$)		ECC60	809 ± 311	(405–1675)	755 ± 314	(93–1418)	0.19
Dt100 HRR	TIME***	ECC30	461 ± 208	(191–988)	628 ± 275	(134–1129)	0.59
($N·s^{-1}$)		ECC60	496 ± 287	(104–1219)	594 ± 341	(57–1306)	0.35
Dt10 amplitude	TIME***	ECC30	58.9 ± 17.5	(28.4–96.9)	36.3 ± 17.7	(10.0–65.3)	1.23
(N·m)	CADENCE*	ECC60	62.9 ± 21.9	(28.9–101.9)	42.8 ± 15.8	(11.9–69.2)	1.09
Dt10 Ct (ms)	TIME***	ECC30	159.7 ± 6.6	(145.0–172.0)	146.8 ± 13.3	(124.5–172.5)	1.02
		ECC60	159.2 ± 11.4	(137.0–179.5)	147.1 ± 16.9	(105.0–173.5)	0.96
Dt10 RTD	TIME***	ECC30	369 ± 107	(170–567)	243 ± 107	(75–410)	1.07
($N·s^{-1}$)	CADENCE*	ECC60	397 ± 148	(192–740)	289 ± 104	(113–478)	0.92
Dt10 HRR	TIME*	ECC30	530 ± 230	(150–1074)	432 ± 259	(98–971)	0.37
($N·s^{-1}$)		ECC60	588 ± 309	(139–1396)	521 ± 256	(65–1075)	0.25
Tw	TIME***	ECC30	37.4 ± 11.3	(18.07–60.71)	29.6 ± 12.8	(10.1–51.8)	0.59
(N·m)		ECC60	39.1 ± 15.5	(18.70–71.90)	31.8 ± 12.7	(7.2–53.3)	0.55
Tw Ct (ms)	TIME*	ECC30	72.7 ± 8.5	(58.5–86.0)	64.6 ± 11.0	(52.0–76.0)	0.55
		ECC60	72.3 ± 17.1	(41.5–88.5)	66.4 ± 19.9	(43.0–85.5)	0.40
Tw-RTD	No effect	ECC30	520 ± 157	(227–788)	457 ± 165	(148–741)	0.33
($N·s^{-1}$)		ECC60	562 ± 232	(164–1048)	512 ± 214	(52–931)	0.26
Tw-HRR	No effect	ECC30	371 ± 183	(42–726)	354 ± 172	(88–733)	0.07
($N·s^{-1}$)		ECC60	372 ± 225	(72–998)	475 ± 289	(54–1008)	0.47
VLM_{MAX} duration (ms)	No effect	ECC30	18.2 ± 4.7	(12.5–27.0)	17 ± 5.1	(10.5–26.5)	0.23
		ECC60	18.6 ± 4.9	(10.5–26.0)	17.8 ± 5.7	(8.0–27.0)	0.16
VLM_{MAX} amplitude (mV)	No effect	ECC30	1.68 ± 0.69	(0.49–2.96)	1.37 ± 0.65	(0.29–2.46)	0.07
		ECC60	1.26 ± 0.77	(0.30–2.74)	1.10 ± 0.67	(0.16–2.52)	0.21
VL	No effect	ECC30	8 ± 3	(2–13)	6 ± 3	(1–11)	0.07
M_{MAX} area (mV·ms)		ECC60	6 ± 4	(1–16)	5 ± 3	(2–10)	0.21
RF	TIME***	ECC30	19.3 ± 2.3	(16.00–23.00)	18.2 ± 2.4	(14.00–22.50)	0.41
M_{MAX} duration (ms)		ECC60	20 ± 3.5	(14.00–28.00)	18.9 ± 3.4	(12.50–26.50)	0.38
RF	No effect	ECC30	1.10 ± 0.47	(0.37–2.05)	1.06 ± 0.46	(0.31–1.89)	0.07
M_{MAX} amplitude (mV)		ECC60	1.16 ± 0.53	(0.25–2.13)	1.04 ± 0.51	(0.25–1.95)	0.24
RF	No effect	ECC30	5 ± 2	(2–9)	5 ± 3	(1–13)	0.15
M_{MAX} area (mV·ms)		ECC60	6 ± 2	(1–9)	5 ± 3	(1–10)	0.17

*indicates an effect on the corresponding factor of the two-way repeated-measures ANOVA. For every symbol, one sign indicates P < 0.05 and three signs indicate P < 0.001. The effect size relates the difference between PRE and POST.

30 rpm compared with 60 rpm. Indeed, both VL and RF muscles showed a main effect of CADENCE, highlighting greater neuromuscular recruitment (number of motor units and/or discharge frequency) at 30 rpm compared with 60 rpm (11). Furthermore, the rise in oxygen consumption seems to be partly associated with the increase in EMG activity of the VL muscle only. Finally, even if the knee extensor muscles are known to absorb the larger part of the power output during ECC cycling (1), the greater increase in oxygen consumption at 30 rpm throughout the duration of exercise could be mediated by a progressive contribution of other muscles.

Perception of effort is known to be mediated by central feedback (the corollary discharge associated with the efference copy of the central motor drive) and strongly associated with respiratory frequency during concentric cycling (20). Present results showed that, overall, perception of effort was moderately correlated with respiratory frequency ($r_{rm} = 0.38$, $P < 0.001$) and strongly correlated with EMG activity of VL muscle ($r_{rm} = 0.55$, $P < 0.001$). When conditions were assessed separately, perception of effort was largely correlated with respiratory frequency at 30 rpm but not at 60 rpm. This absence of correlation at 60 rpm has already been observed in a previous study using similar power output (4). In parallel, for both cadences assessed separately, the perception of effort was strongly correlated EMG activity of VL muscle (all $r_{rm} = 0.59$, all $P < 0.001$) (21). Because the EMG activity of the VL muscle is largely lower at 60 than at 30 rpm, it seems that the lower force applied to the pedals may contributes to the lower perception of effort during ECC cycling at 60 rpm. Conversely, the greatest respiratory frequency at 30 than 60 rpm, as well as the presence of a large correlation between perception of effort and respiratory frequency at 30 rpm but not at 60 rpm, might indicate that respiratory frequency has a significant contribution to perception of effort at lowest the cadence. Lechauve et al. (22) also showed a concomitant increase in respiratory frequency with power output during an ECC cycling incremental test. It was explained by a greater requirement of trunk stabilization, highlighted by greater muscle activity of the biceps brachial muscle at high intensity and lower cadence, which could be associated with a limitation for rib cage expansion. It is possible that this needs to keep the trunk fixed at 30 rpm, which was considered when participants reported their perception of effort, although it was asked to the participants to focus on their difficulty to breath and hold the movement of the pedals. Indeed, the strong correlation is observed between the perception of effort and the EMG activation of the BB at 30 rpm only. All the points mentioned previously may partly explain the greater perception of effort when cycling at 30 rpm compared with 60 rpm.

As previously demonstrated, muscle pain could be dissociated from perception of effort (23) because the underlying phenomena are different (24). However, although pain does not directly affect the central processes associated with the perception of effort, it can nevertheless impact both the ability to generate force and patients' adherence to exercise (25). Here muscle pain, referred as "how it hurt in the quadriceps muscles," increased during cycling without difference between cadences. However, it is possible that the reasons are not the same for both cadences. Indeed, participants' oral comments during the 30-rpm condition sounded more like pain associated with muscle compression. Those informal exchanges let us think that mechanical muscle tension induced by the lowest cadence, thus high level of torque upon pedals, could activate type III–IV mechanosensitive nociceptive afferents known to contribute to the onset of acute muscle pain (23). On the other hand, participants expressed difficulty in maintaining an exact cadence of 60 rpm but suggested that it would probably be easier around 55 rpm. A cadence slightly less than 60 rpm corresponds to the cadence estimated to minimize muscle activation in our previous study (55.9 ± 9.3 rpm) (11). Indeed, although instruction was to focus on pain, it is possible that the participants, despite being familiarized during the first session, did not fully dissociate exercise-related sensations such as muscle pain with discomfort or pain from body parts such as foot and knee (26).

Neuromuscular alterations after cycling exercise. Pedaling cadence had no effect on the decrease in maximal voluntary isometric contraction torque. MVC loss immediately after exercise was in line (≈−18%) with results obtained with a similar experimental design (i.e., 30 min at 60% PPO and 60 rpm) (4,6). This reduction was not accompanied by neural alterations (VAL, EMG_{MAX} for VL and RF muscles), regardless of the cadence. Our results do not agree with all the findings of one study mentioned previously. Indeed, although Clos et al. (2022) showed a decrease in VAL (≈−2.2%) after ECC cycling at 60 rpm, our data did not reveal a significant decline, which could be explained by a high interindividual variability. Conversely, in both studies, maximal central drive (EMG_{MAX} during MVC) was not altered whether cycling exercise was performed at 30 or 60 rpm. Clos et al. (2022) assumed that the decrease in VAL was due to the discharge of nociceptive afferents III–IV (known to inhibit the motor control of knee extensor muscles and alter the spinal path) (27) because the participants reported muscular pain during walking immediately after exercise. Our longer familiarization period than Clos et al. (2022) (30 vs 15 min) may have induced a repeated bout effect and, consequently, more significant neural adaptations after the first session of ECC cycling (28). This repeated bout effect could have limited the impact of muscular pain during the second exposure, by reducing the discharge of afferents.

Muscle activation of both VL and RF muscles demonstrated an overall effect of cadence, highlighting greater neuromuscular recruitment (number of motor units and/or discharge frequency) at 30 rpm compared with 60 rpm (11). However, despite stable power output, an increased EMG of the VL muscle with no difference between cadences during exercise evidenced a reduced muscle efficiency, regardless of cadence. In addition, muscular function was similarly affected by ECC cycling for both cadences without a cadence effect. Along with the decrease in MVC torque, the present study showed a reduction of the amplitude of electrically evoked torque (Dt100, Dt10, and Tw) after exercise without difference between conditions. In addition, sarcolemmal excitability (i.e., M-wave) was moderately impacted by exercise, again without the influence of experimental conditions as suggested by the decrease in M_{MAX} duration of the RF muscle, whereas an increase in EMG was only shown for the VL muscle during the exercise. This difference between the muscles can be explained by their functional differences. Although the VL muscle is monoarticular and absorbs the majority of the power, the RF muscle, being biarticular, is more involved in force orientation. The alterations observed during an ECC exercise (i.e., an increased in EMG signal of the VL muscle only) do not always reflect those observed with neuromuscular tests performed after exercise in isometric condition. Indeed, it has been shown that a maximal voluntary isometric contraction does not highlight the specific decreased in functional capacity caused by a fatiguing exercise in concentric and ECC modes (29). Finally, the reduction in the low-to-high frequency stimulation response (Dt10/Dt100 ratio) suggested an alteration in excitation–contraction coupling known as specific ECC-induced muscular alteration affecting Ca^{2+} release in the sarcoplasmic reticulum (30). These results confirm that locomotor ECC exercise notably alters excitation–contraction coupling (4,31,32). Furthermore, muscle soreness reported in the days after experimental sessions was indirect evidence of histological damage.

CONCLUSIONS

When performed at the same power output, ECC cycling exercise at 30 rpm elicited a greater perception of effort, muscle activation, and cardiorespiratory demands than pedaling at 60 rpm. However, performance fatigability (i.e., MVC torque decline) and neuromuscular alteration (e.g., excitation–contraction coupling alteration) after exercise were not influenced by cycling cadence, suggesting a dissociation between modulations during exercise and exercise-induced neuromuscular alterations. In rehabilitation, clinicians need to bear in mind that ECC cycling performed at low cadence is perceived more difficult and induces greater cardiovascular demands. However, if a lack of coordination forces patients to use a low cadence (8,9) at the start of the training period, the intensity of the exercise should be reduced to avoid generating a too high perception of effort, which would hinder patients in their physical activity. Then, because used higher cadenced should allow to absorb greater power output (Green et al., 2018) and cadence about 60 rpm should lead to the lower cardiorespiratory demands, rehabilitation program could propose to gradually increase cadence from 30 to 60 rpm with an increase in power output.

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APPLIED SCIENCES

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